

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

1. Network for a 20-Person Startup

A **star topology** is the most suitable choice for a 20-person startup because every device is connected to a central switch. A **24-port managed Layer 2/Layer 3 switch** can connect computers, printers, VoIP phones, and other devices while leaving additional ports for future expansion.

Each device can be connected using **Cat6 UTP cables**, which support Gigabit Ethernet over distances of up to 100 meters. The central switch can be connected to a **router/firewall** for Internet access, DHCP, NAT, and security. A wireless access point can also be connected to the switch to provide Wi-Fi.

The major advantage is **easy fault isolation**. If one cable or workstation fails, only that device is affected and the remaining users continue working. Therefore, star topology provides good **manageability, scalability, reliability, and troubleshooting** for a small startup.

2. Network for a Bank Requiring Near-Zero Downtime

A **partial mesh topology** is appropriate for a bank because banking systems require high availability and minimal downtime. In a mesh network, important devices have multiple connections, so if one link fails, traffic can use another available path.

At the core, routers, switches, and firewalls can have **redundant connections**. Distribution switches can connect to two core switches so that failure of one core device does not interrupt network communication. Dual enterprise routers can provide redundant Internet connections.

Fiber-optic cables are suitable for connections between important network devices because they provide high bandwidth, low latency, long-distance communication, and immunity to electromagnetic interference.

Protocols such as **OSPF or BGP** can automatically select alternative paths when a connection fails. Redundant power supplies, UPS systems, and RAID storage can further improve availability.

3. Network for a Small Classroom Lab

For a small classroom with a limited budget, a **bus topology** can be considered because all computers share a common communication medium. Traditional bus networks use a single coaxial backbone cable with **T-connectors, BNC connectors, and terminators** at both ends.

The main advantage is that it requires less networking hardware compared with a traditional star topology. It can therefore reduce the initial cost.

However, bus topology has important disadvantages. A failure in the main backbone cable can affect the entire network. Network traffic can also cause **collisions and performance problems**, particularly when many computers communicate simultaneously.

For a modern classroom, however, a small **unmanaged Ethernet switch with Cat5e/Cat6 cables** is generally more practical and reliable than a true coaxial bus network.

4. Network for a Large University Campus

A **hybrid topology** is the most suitable design for a large university because thousands of users are distributed across multiple buildings. Different topologies can be combined according to the requirements of each area.

At the campus backbone, a **fiber-based ring or partial mesh** can provide redundancy. Each building can have a distribution switch with **dual fiber connections** to the campus core.

Inside individual buildings, a **star topology** can be used. The distribution switch connects to floor-level access switches, and the access switches connect computers, printers, and other devices using Cat6 cables.

A three-layer hierarchical architecture can be used:

Core → Distribution → Access

Wireless access points can be connected to access switches using **Power over Ethernet (PoE)**.

This design provides **high scalability, redundancy, easy management, and efficient expansion**. A new building can be added by connecting its distribution switch to the existing campus backbone.

5. Network for a Company with Multiple Departments

A **hierarchical network with VLANs** is suitable for a company containing departments such as HR, Finance, Sales, and IT.

Each department can be connected to a dedicated **Layer 2 access switch**. Devices within a department can communicate through the switch. Separate **VLANs** can be created for each department.

For example:

- **VLAN 10 → HR**
- **VLAN 20 → Finance**
- **VLAN 30 → Sales**
- **VLAN 40 → IT**

A **Layer 3 switch or router** performs **inter-VLAN routing**, allowing authorized communication between departments. A firewall/router connects the internal network to the Internet and provides security.

IP addressing and subnetting can be used to organize each department's network efficiently. Redundant links can also be added to improve reliability.

The use of VLANs reduces unnecessary broadcast traffic and improves **security, performance, scalability, and network management**.